

Strategic Blueprint: From Prototype to Production-Ready LLM Features

1. Mission Brief

You are the Lead AI Architect. Your goal isn't just to "call an API"—it's to architect a robust, scalable LLM system that solves a high-stakes engineering bottleneck. You will design, build, and defend a solution that demonstrates Architectural Thinking and Production Readiness.

2. The innovation journey (Sprint Phases)

Phase 1: The High-Value Target

- Action: Don't pick an easy problem; pick a painful one.
- Examples: An AI that performs *Root Cause Analysis* on production incidents, or a *RAG-based agent* that generates PR descriptions from Jira tickets and code diffs.
- Goal: Define the ROI (Return on Investment). Why should the company spend \$0.05 per call on this?

Phase 2: System Architecture (The Blueprint)

- Action: Design your "Orchestration Layer." Will you use LCEL (LangChain Expression Language)? An Agentic workflow with Tool-calling?
- Engineering Requirement: You must include a diagram showing the data flow: *User -> Embedding -> Vector Store -> LLM -> Output Parser*.

Phase 3: Prototype & Iteration (The Lab)

- Action: Build the core logic.
- The "Pro" Touch: Implement at least one advanced optimization technique, such as:
 - Parent Document Retrieval (for better context).
 - Self-Querying (to handle metadata).
 - Contextual Compression (to save tokens and reduce latency).

Phase 4: The Reality Check (Trade-offs & Production)

- Action: Defend your decisions.
- Critical Thinking: Why did you choose *gpt-4o-mini* over *claude-3.5-sonnet*? How are you handling Prompt Injection? What is your fallback if the Vector Store returns 0 results?

3. Your Architect (Deliverables)

To pass the Engineering Review, you must submit to Moodle:

1.  The Technical Design Document (TDD):
 - A 5-8 page PDF. Focus on the "Why" more than the "What."
 - Include your Architecture Diagram.
2.  The Repository (Codebase):
 - A clean, documented GitHub repo or ZIP.
 - Must include a [README.md](#) with instructions on how to run the prototype.
3.  Evaluation Evidence:
 - Screenshots of LangSmith traces or manual logs showing how you fixed a hallucination or improved a prompt.

4. Evaluation Criteria

Score	Criteria	The "Excellent" Standard
20%	Architecture	System is modular, using RAG or Agents effectively.
20%	Implementation	Prototype works and handles edge cases (e.g., empty inputs).
15%	Iteration	Evidence that you tested, found a flaw, and improved it.
10%	Prod-Readiness	Concrete plan for monitoring, security, and cost control.
35%	Problem & Logic	Clear business value and sound technical reasoning.